Teamwork Agreement Template

For the MathWorks Classroom Challenge Projects

Project: Battery Profile

Team Name: Team3

Instructor: Dr. Chintakunta and Dr. Guerrero

Program/Class: Engineering Pathway Program 2026

Members and Roles

1. Cris Cervantes Gonzalez - Analysis/Validation Lead
2. Soliana Hadera - Project Manager (PM)
3. Vannara Net - Modeling Lead
4. Aren Safarian - Documentation & Visualization Lead

   All will assume responsibility regarding quality assurance and reproducibility.

1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts (e.g. plots, code, and visualizations).

2) Roles & Responsibilities

- Project Manager (PM): Schedules meetings; maintains timeline; ensures tasks are tracked and disagreements are resolved.
- Modeling Lead: Owns core MATLAB models/simulations
- Analysis/Validation Lead: Performs hand calculations and cross-checks simulation outputs
- Documentation & Visualization Lead: Compiles report (with figures/tables), prepares slides (if needed), and builds visual artifacts (e.g. animations or plots)
- Quality Assurance & Reproducibility Lead: Enforces coherent style across artifacts, ensures work is not overwritten across different versions, ensures data quality, tests/re-runs codes to ensure robustness and reproducibility

3) Communication Norms

- Primary channel: Discord

- Secondary channel: E-mail
  crise.cervantes@gmail.com
  vannara091@gmail.com
  solianabhadera@gmail.com
  aren.safarian@gmail.com

- Response time: 24 hours on weekdays and 48 hours on weekends (e.g. within 24 hours on weekdays; within 48 hours on weekends)

- Meeting cadence: TBD

- Decision records: PM logs major decisions in a running document (Minutes)

4) Collaboration & Tools

- Version control: github

- Data management: All datasets, parameters, and assumptions documented in

  README.md

File naming & style: descriptive names

- Backups: files sent to discord groupchat

5) Quality Standards (Definition of 'Done')

Requirements meet project description, code is reproducible (runs from the README.md file, all specified figures regenerate, at least one teammate has tested edge cases, all assumptions, units, and references are documented in report).

Team members are able to pinpoint their contribution, as well as demonstrate understanding of project.

6) Timeline & Milestones

- Week 1: Problem framing, literature/background review, dataset/parameters,
- Week 2: First pass models (hand calculations + baseline MATLAB scripts); initial plots/visuals.
- Week 3: Validation and quality assurance
- Week 4: Report, slides, presentation rehearsal, repository cleanup

7) Decision-Making & Conflict Resolution

- Default: Consensus after discussion; PM facilitates
- If blocked >48 hrs: Time-limited debate, majority vote, PM documents decision
- Technical disagreements: Evidence-based comparison before vote

- Escalation: If unresolved, consult instructor/mentor with a concise memo.

8) Workload, Accountability & Attendance

- Task board: To-Do / Doing / Review / Done with owners and due dates documented in README.md

- Accountability: Missed deadline → recovery plan within 24 hrs; repeated misses trigger role redistribution.

- Attendance: If you'll miss a meeting, give 24-hour notice and share updates asynchronously.

10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

- A fitted charging-curve model for a real lithium-ion battery dataset, including a graph of the data with the fitted curve and goodness-of-fit statistics
- Plots for: voltage vs. time, current vs. time, and power vs. time
- Analytical results (numerical values or derivatives), presented in a summary table
    - Numerical integration results showing total energy delivered during charging.
    - Time required to reach 80% and 100% charge.
    - Rate-of-change analysis (derivatives at key intervals).
    - Estimates of resistive energy loss.

12) Agreement & Signatures

Name & Signature: Cristian Cervantes Gonzalez Date: 7/8/2026

Name & Signature: Soliana Hadera Date: 07/08/2026

Name & Signature: Vannara Net Date: 07/08/2026

Name & Signature: Aren Safarian Date: 7/09/2026